CSCE 4350 Project

By Group 7:
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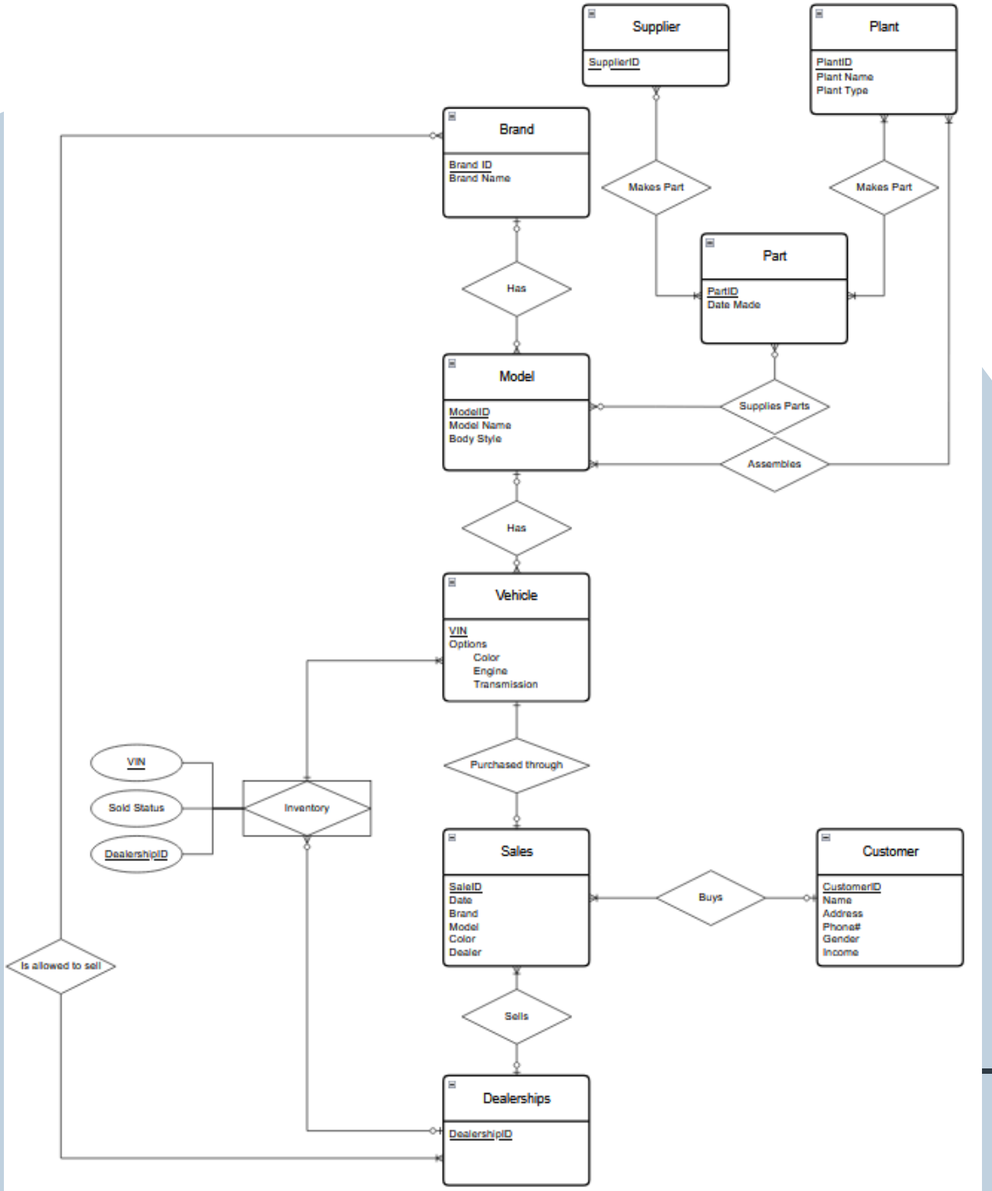
ER DIAGRAM



The diagram imitates the process of selling a vehicle, from sourcing parts, to assembly, to dealership, to inventory, and to final sale to the customer. The diagram tracks key entities, like vehicles, brands, models, parts, suppliers, plants, dealers, customers, and sales, with their respective primary keys, to demonstrate how this data interacts across the companies manufacturing plants and the dealership operations. An associative entity was added for inventory, as inventory isn't a relationship, but it's not an entity either, as neither vehicle or dealership can be independent with inventory.

DEVELOPMENT SCHEDULE

Development Schedule			
Task	**Expected Completion Date**	**Completed**	**Description**
Relational Model			
Create Relations in MySQL database	12-Apr	Done	Generate the schema for the SQL Database
Generate Testing Data (Populate Relations)		Done	
Create code to generate Testing data	14-Apr	Done	Create Sample data that can be imported into the database
Create Queries		Done	
Sales Trends by Year, Month, Week -> Gender + Income	17-Apr	Done	Generate the SQL queries
Identify what cars contain parts by supplier made between dates. Locate VIN and customer			
Top 2 brands by sale price			
Top 2 brands by units sold			
Best months for convertibles			
Dealerships that keep inventory for the longest average time			
Generate Interfaces		Done	
Generate UI	18-Apr	Done	Generate a simple button UI for each of these portals
Customer			
Sales			
Main Menu			
Customer Portal			
Marketing Portal			
Vehicle locator		Done	
Local and nearby implementation	20-Apr	Done	Create the Vehical locator which has distance and option matching
Options matching			
Integrate Concurrency	25-Apr	Done	
Java Implementation	22-Apr	Done	
Testing	26-Apr	Done	
Update ER Diagram	23-Apr	Done	
Create README File	28-Apr	Done	
Submit Files	**30-Apr**		

The development schedule shows our timeline for when we expect to complete each major aspect of the project. We split the sections into inquiries, concurrency, and interfaces to make the project more manageable as well as to ensure a good project flow. Each task was pulled directly from the project outline and a completion data was assigned based on the expected amount of work that needed to be put in.

RELATIONAL SCHEMA PT 1

```sql
--Relation Schema

CREATE TABLE Brand (
    brand_id SERIAL PRIMARY KEY,
    name VARCHAR(50) UNIQUE NOT NULL
);

CREATE TABLE Model (
    model_id SERIAL PRIMARY KEY,
    brand_id INT,
    name VARCHAR(50),
    FOREIGN KEY (brand_id) REFERENCES Brand(brand_id)
);

CREATE TABLE BodyStyle (
    body_style_id SERIAL PRIMARY KEY,
    description VARCHAR(50)
);

CREATE TABLE Color (
    color_id SERIAL PRIMARY KEY,
    name VARCHAR(30)
);

CREATE TABLE Engine (
    engine_id SERIAL PRIMARY KEY,
    type VARCHAR(50)
);

CREATE TABLE Transmission (
    transmission_id SERIAL PRIMARY KEY,
    type VARCHAR(50)
);
```

```sql
CREATE TABLE Vehicle (
    VIN VARCHAR(17) PRIMARY KEY,
    model_id INT,
    body_style_id INT,
    color_id INT,
    engine_id INT,
    transmission_id INT,
    year INT,
    FOREIGN KEY (model_id) REFERENCES Model(model_id),
    FOREIGN KEY (body_style_id) REFERENCES BodyStyle(body_style_id),
    FOREIGN KEY (color_id) REFERENCES Color(color_id),
    FOREIGN KEY (engine_id) REFERENCES Engine(engine_id),
    FOREIGN KEY (transmission_id) REFERENCES Transmission(transmission_id)
);

CREATE TABLE Dealer (
    dealer_id SERIAL PRIMARY KEY,
    name VARCHAR(100),
    address VARCHAR(200),
    phone VARCHAR(20)
);

CREATE TABLE Customer (
    customer_id SERIAL PRIMARY KEY,
    name VARCHAR(100),
    address VARCHAR(200),
    phone VARCHAR(20),
    gender CHAR(1),
    income DECIMAL(12,2)
);

CREATE TABLE Inventory (
    inventory_id SERIAL PRIMARY KEY,
    VIN VARCHAR(17),
    dealer_id INT,
    status VARCHAR(20),
    FOREIGN KEY (VIN) REFERENCES Vehicle(VIN),
    FOREIGN KEY (dealer_id) REFERENCES Dealer(dealer_id)
);
```

```sql
CREATE TABLE Sales (
    sale_id SERIAL PRIMARY KEY,
    VIN VARCHAR(17),
    dealer_id INT,
    customer_id INT,
    sale_date DATE,
    price DECIMAL(10,2),
    FOREIGN KEY (VIN) REFERENCES Vehicle(VIN),
    FOREIGN KEY (dealer_id) REFERENCES Dealer(dealer_id),
    FOREIGN KEY (customer_id) REFERENCES Customer(customer_id)
);

CREATE TABLE Supplier (
    supplier_id SERIAL PRIMARY KEY,
    name VARCHAR(100),
    contact_info VARCHAR(200)
);

CREATE TABLE Part (
    part_id SERIAL PRIMARY KEY,
    name VARCHAR(100)
);

CREATE TABLE Supplier_Part (
    supplier_id INT,
    part_id INT,
    model_id INT,
    PRIMARY KEY (supplier_id, part_id, model_id),
    FOREIGN KEY (supplier_id) REFERENCES Supplier(supplier_id),
    FOREIGN KEY (part_id) REFERENCES Part(part_id),
    FOREIGN KEY (model_id) REFERENCES Model(model_id)
);

CREATE TABLE Plant (
    plant_id SERIAL PRIMARY KEY,
    name VARCHAR(100),
    location VARCHAR(100),
    type VARCHAR(20)
);
```

RELATIONAL SCHEMA PT 2

```sql
CREATE TABLE Plant_Part (
    plant_id INT,
    part_id INT,
    model_id INT,
    PRIMARY KEY (plant_id, part_id, model_id),
    FOREIGN KEY (plant_id) REFERENCES Plant(plant_id),
    FOREIGN KEY (part_id) REFERENCES Part(part_id),
    FOREIGN KEY (model_id) REFERENCES Model(model_id)
);

CREATE TABLE Dealer_Brand (
    dealer_id INT,
    brand_id INT,
    PRIMARY KEY (dealer_id, brand_id),
    FOREIGN KEY (dealer_id) REFERENCES Dealer(dealer_id),
    FOREIGN KEY (brand_id) REFERENCES Brand(brand_id)
);

CREATE TABLE Vehicle_Part (
    VIN VARCHAR(17),
    part_id INT,
    supplier_id INT,
    plant_id INT,
    date_made DATE,
    PRIMARY KEY (VIN, part_id),
    FOREIGN KEY (VIN) REFERENCES Vehicle(VIN),
    FOREIGN KEY (part_id) REFERENCES Part(part_id),
    FOREIGN KEY (supplier_id) REFERENCES Supplier(supplier_id),
    FOREIGN KEY (plant_id) REFERENCES Plant(plant_id)
);
```

This relational schema models an automobile company's interactions, which includes vehicles, brands, models, dealers, customers, sales, inventory, suppliers, and manufacturing plants. There are different relationships that have been implemented using the foreign keys and some junction tables to ensure that it supports analytical queries.

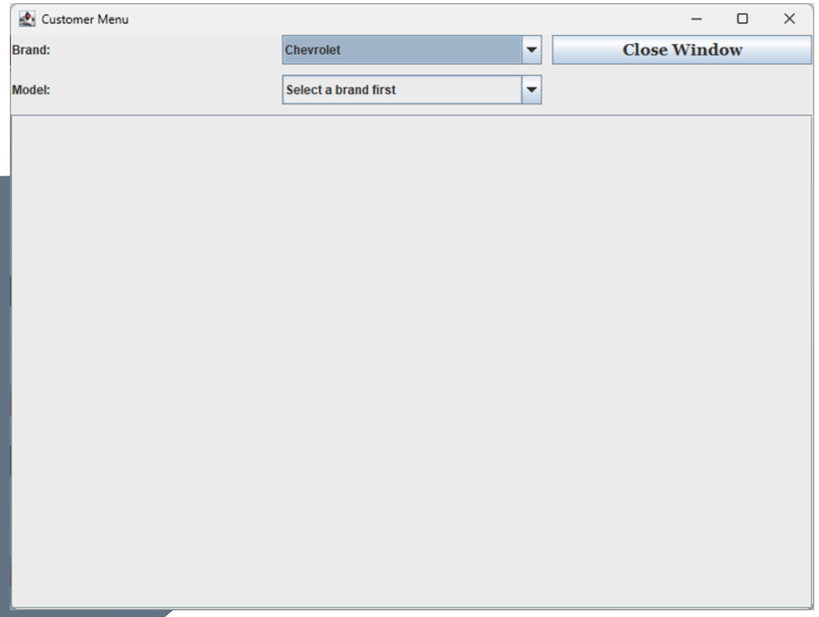
INTERFACE – MAIN MENU

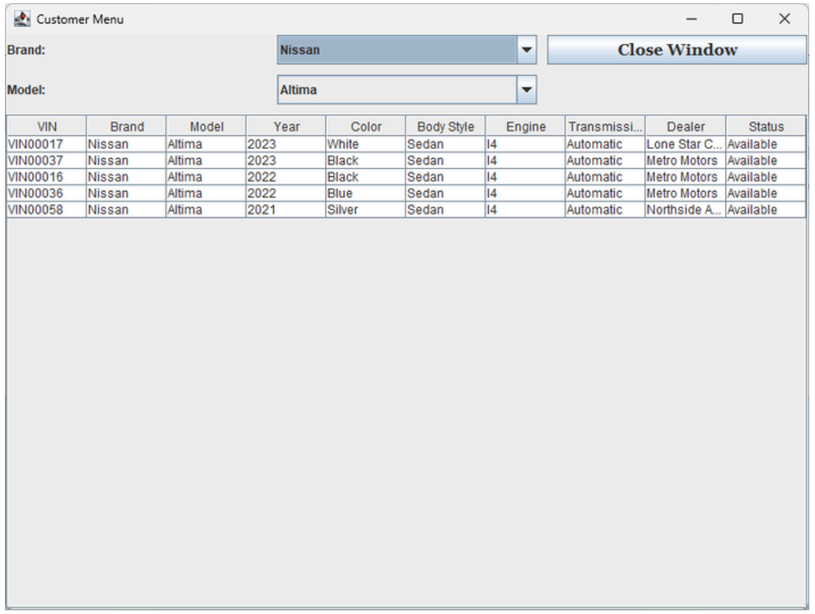
The Main Menu screen shows the main options that users can access.

The team decided that the main options for this project were to be:

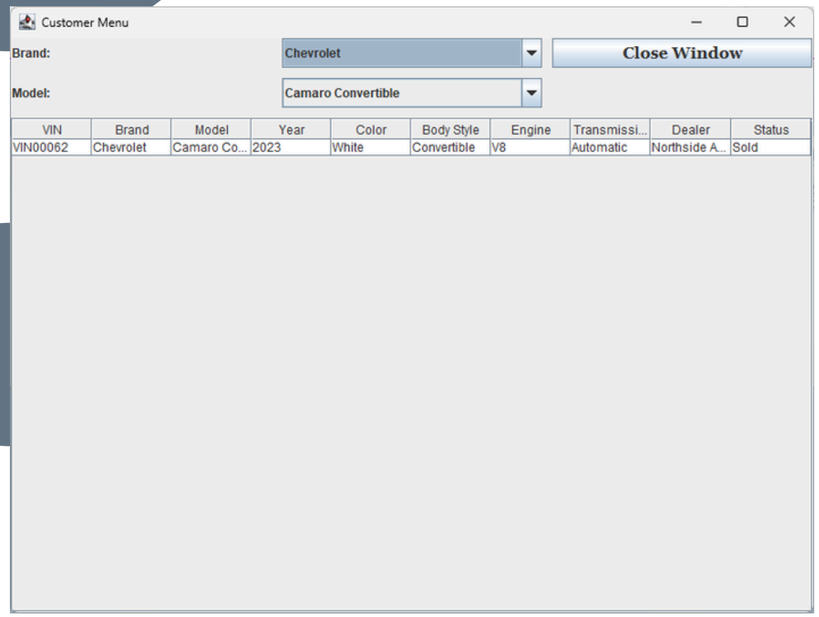
- **Customer menu**, where users can look up the available vehicles in the inventory.
- **Sales menu**, where users can add sales.
- **Reports menu**, showing all the different reports, aligned with the assignment requirements.
- **Vehicle Locator menu**, where users can check the inventory and status of vehicles.
- **Parts Locator menu,** where users can see what sold vehicles have supplier parts.
- **Close Program button**

INTERFACE – CUSTOMER MENU

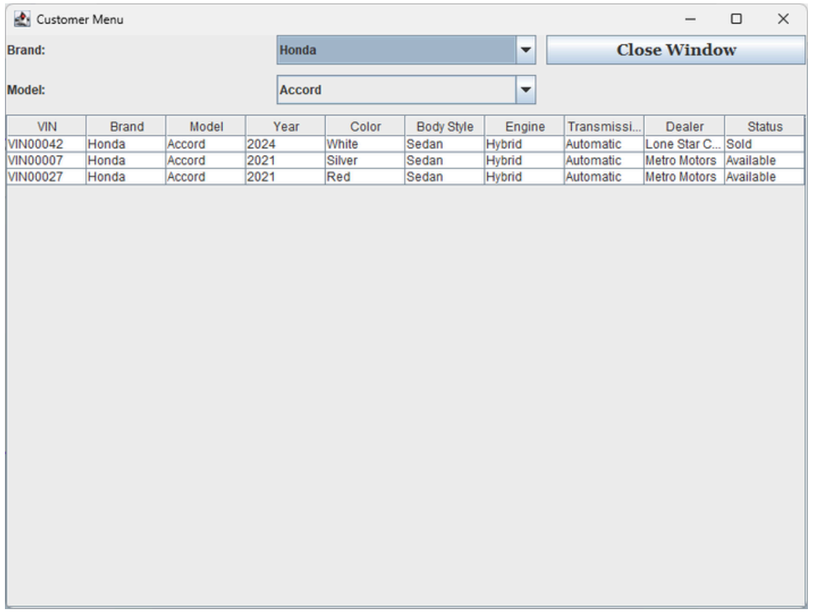




VIN	Brand	Model	Year	Color	Body Style	Engine	Transmissi...	Dealer	Status
VIN00017	Nissan	Altima	2023	White	Sedan	I4	Automatic	Lone Star C...	Available
VIN00037	Nissan	Altima	2023	Black	Sedan	I4	Automatic	Metro Motors	Available
VIN00016	Nissan	Altima	2022	Black	Sedan	I4	Automatic	Metro Motors	Available
VIN00036	Nissan	Altima	2022	Blue	Sedan	I4	Automatic	Metro Motors	Available
VIN00058	Nissan	Altima	2021	Silver	Sedan	I4	Automatic	Northside A...	Available



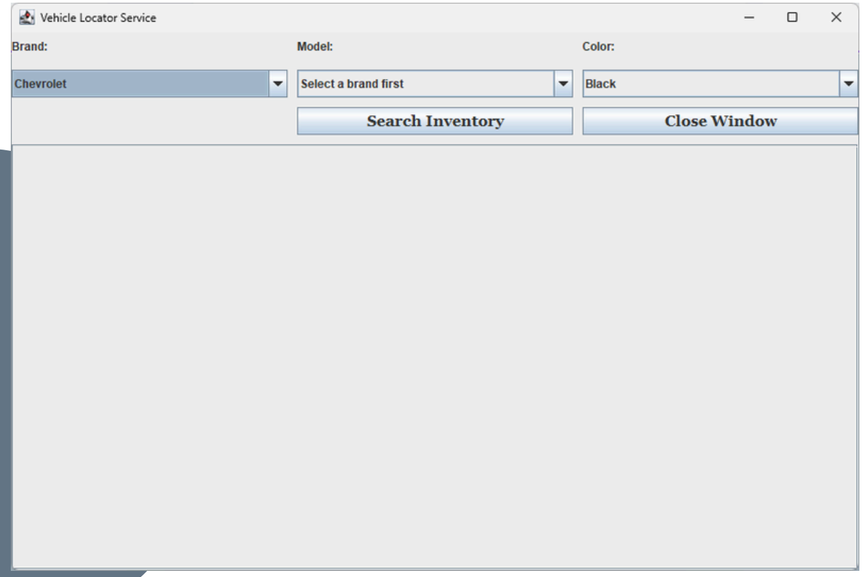
VIN	Brand	Model	Year	Color	Body Style	Engine	Transmissi...	Dealer	Status
VIN00062	Chevrolet	Camaro Co...	2023	White	Convertible	V8	Automatic	Northside A...	Sold

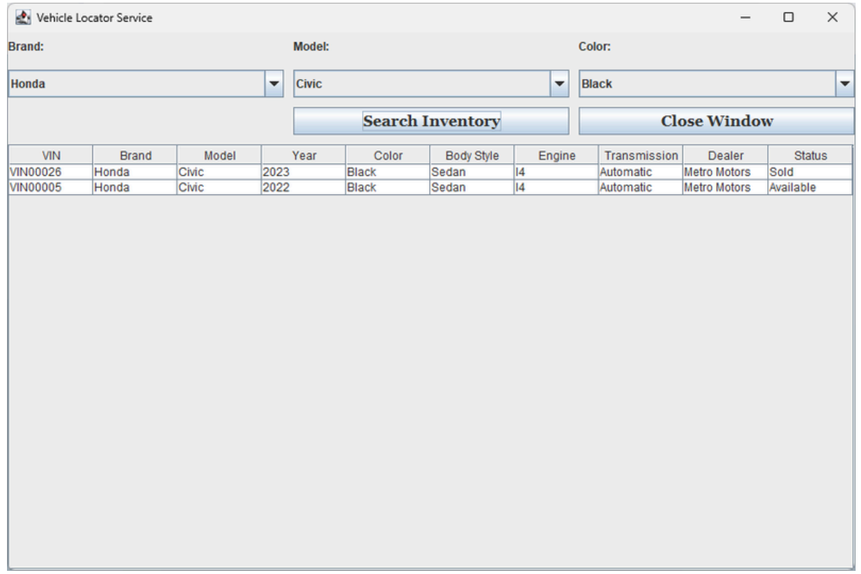


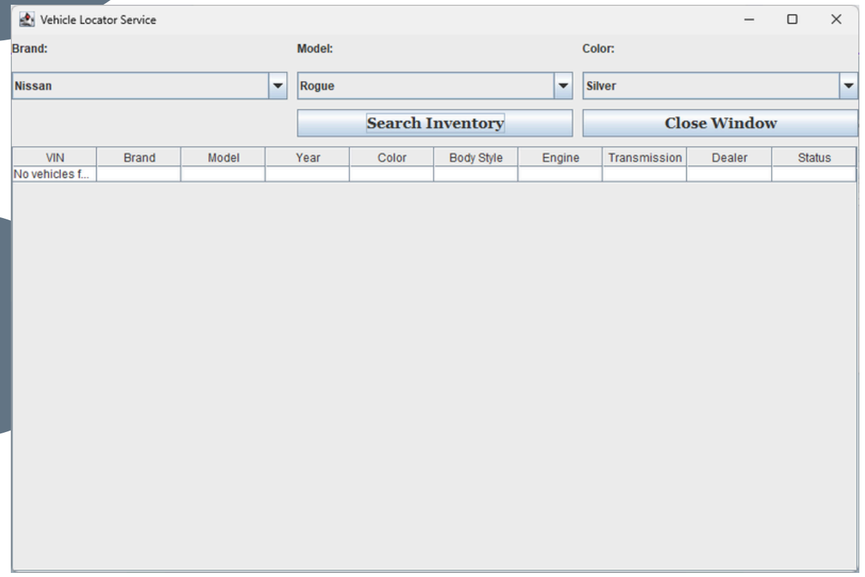
VIN	Brand	Model	Year	Color	Body Style	Engine	Transmissi...	Dealer	Status
VIN00042	Honda	Accord	2024	White	Sedan	Hybrid	Automatic	Lone Star C...	Sold
VIN00007	Honda	Accord	2021	Silver	Sedan	Hybrid	Automatic	Metro Motors	Available
VIN00027	Honda	Accord	2021	Red	Sedan	Hybrid	Automatic	Metro Motors	Available

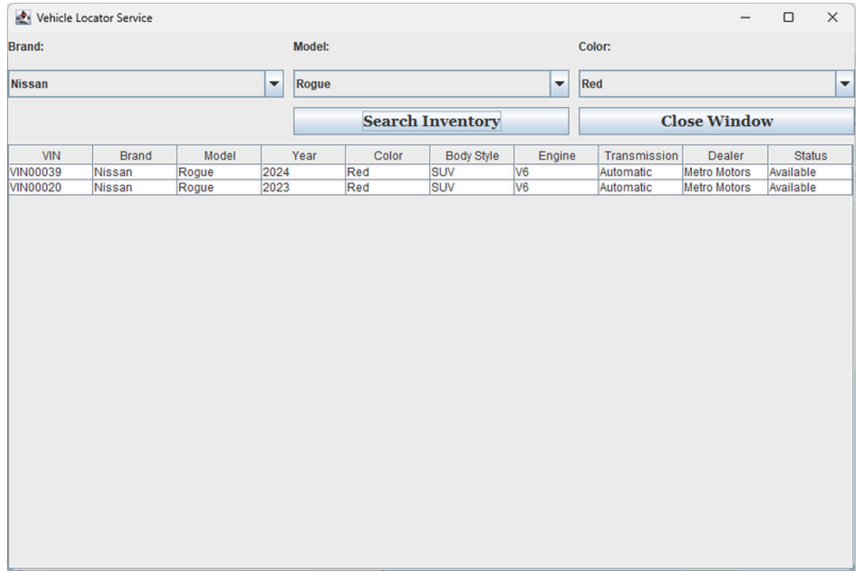
The **Customer Menu** window gives descriptions of the vehicles available in all dealerships' inventories, as well as the status of each vehicle.

INTERFACE – VEHICLE MENU



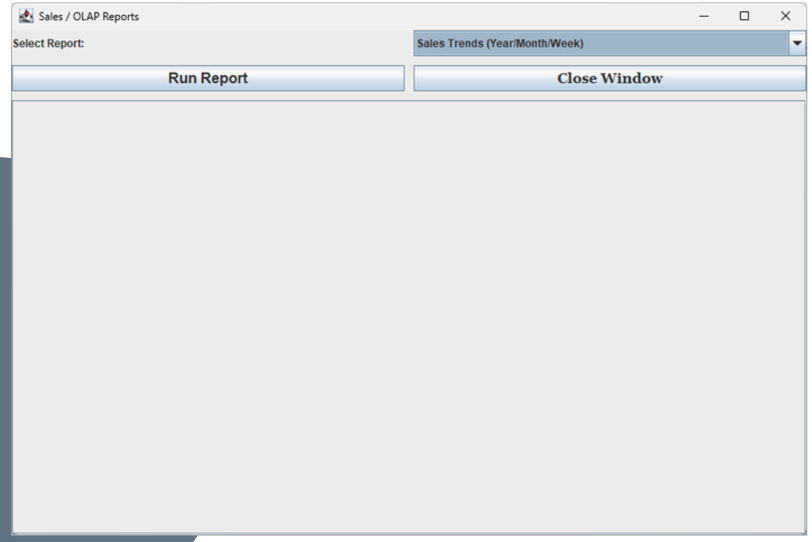


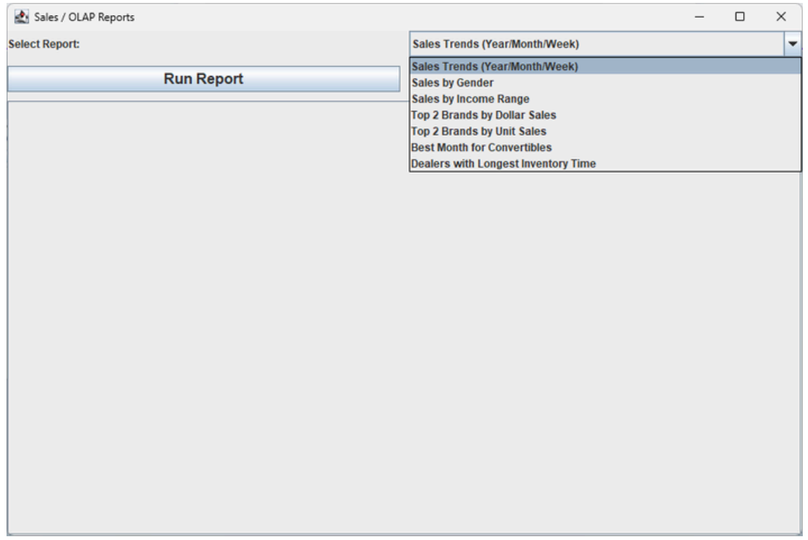


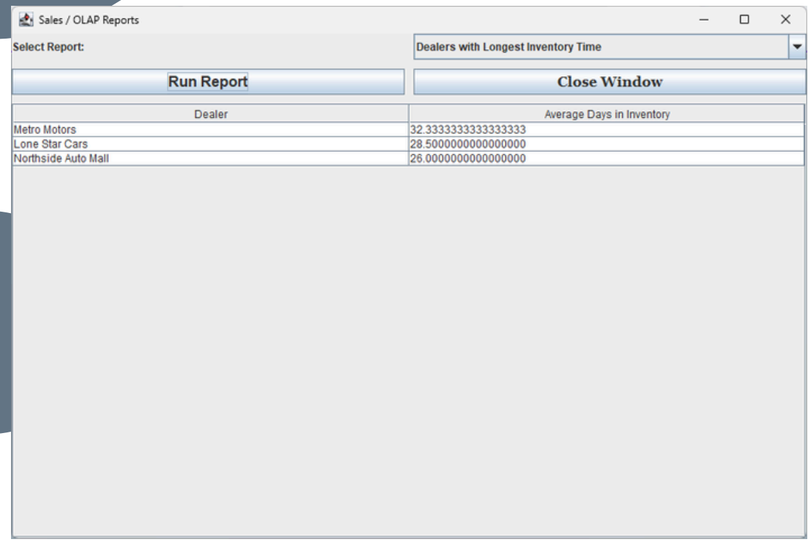


The **Vehicle Menu** is very similar to **Customer Menu**, with the difference of selecting a color for a more specific search. While **Customer Menu** shows all cars, **Vehicle Menu** can also show whether a car exists in the inventory or not.

INTERFACE - REPORTS MENU



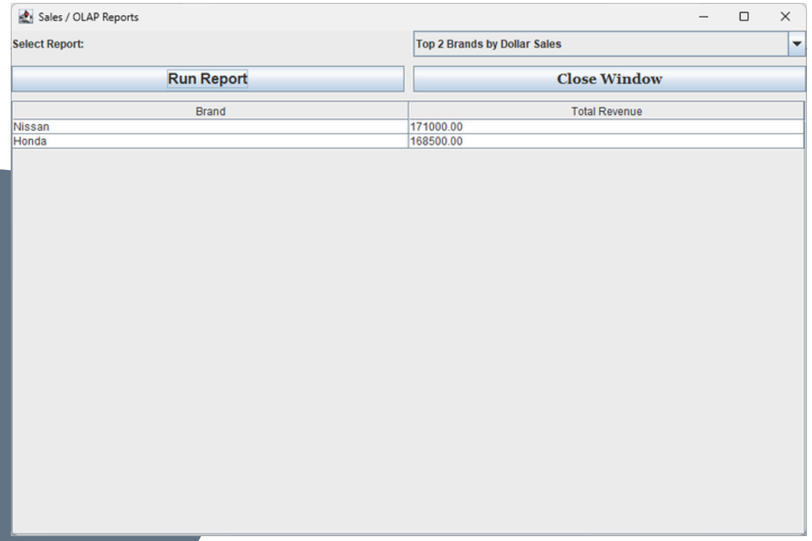




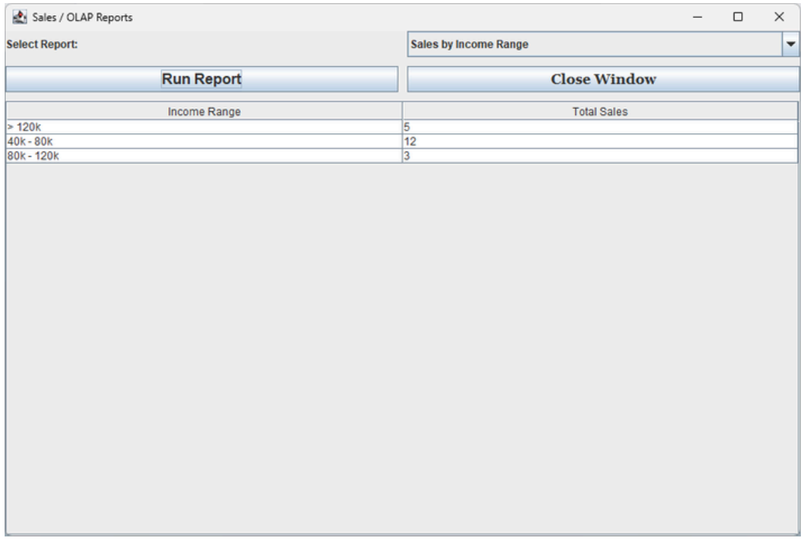
The **Reports Menu** window is where all the information from sales and vehicles in the inventory is shown. The team created the different report options as required by the assignment:

- A report that shows the sales made ordered by year, month and week
- A report showing how many vehicles have been sold to men and how many to women
- Sales by customer's income range
- Top 2 brands sales
- Best month for convertible cars
- Dealers with the longest inventory time (in days)

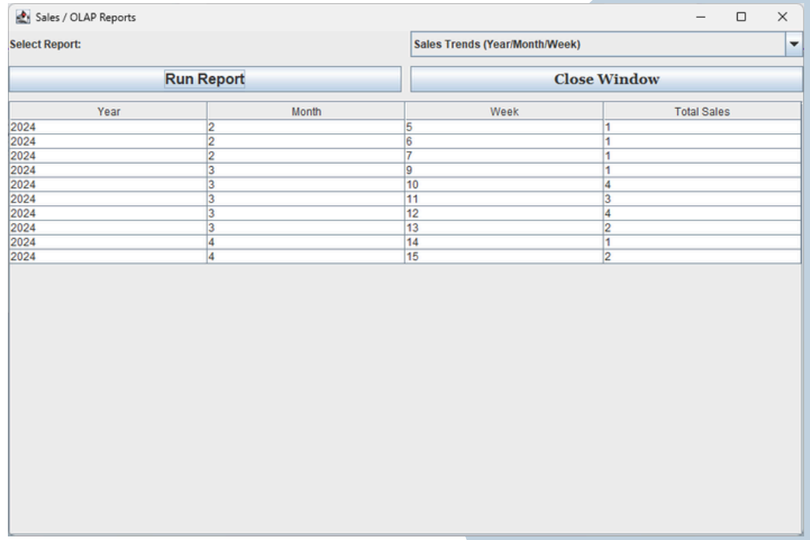
INTERFACE – REPORTS MENU



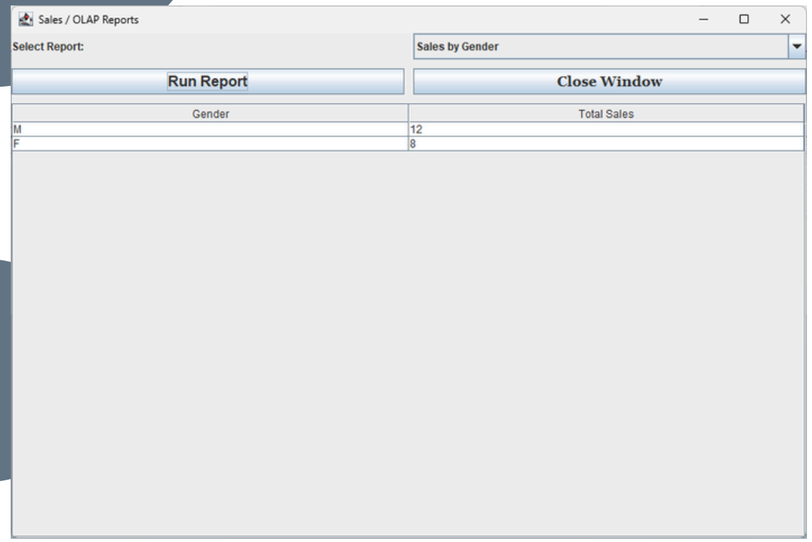
Brand	Total Revenue
Nissan	171000.00
Honda	168500.00



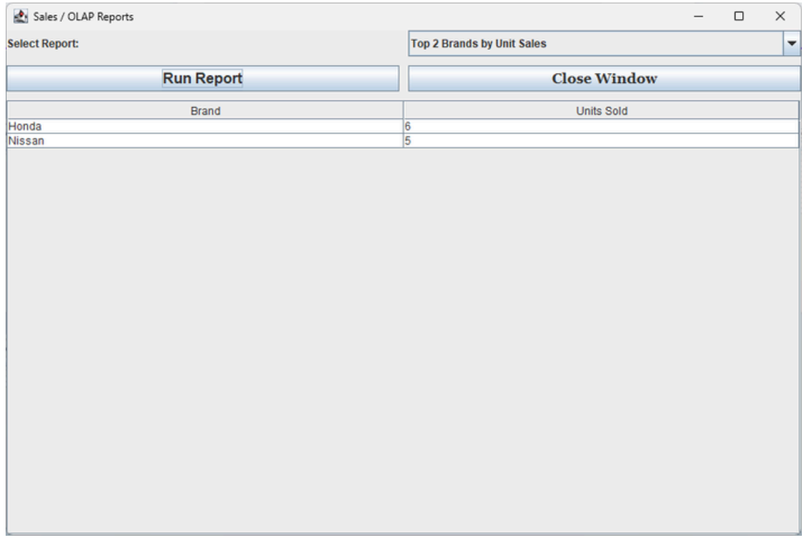
Income Range	Total Sales
> 120k	5
40k - 80k	12
80k - 120k	3



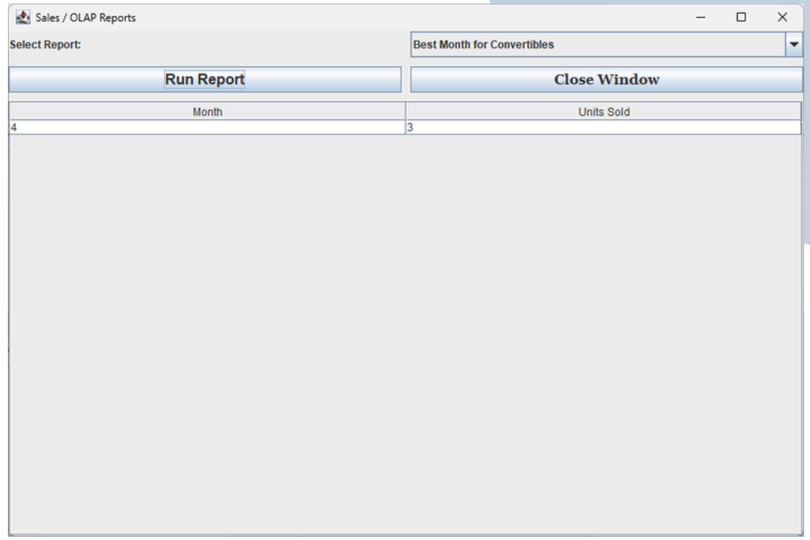
Year	Month	Week	Total Sales
2024	2	5	1
2024	2	6	1
2024	2	7	1
2024	3	9	1
2024	3	10	4
2024	3	11	3
2024	3	12	4
2024	3	13	2
2024	4	14	1
2024	4	15	2



Gender	Total Sales
M	12
F	8

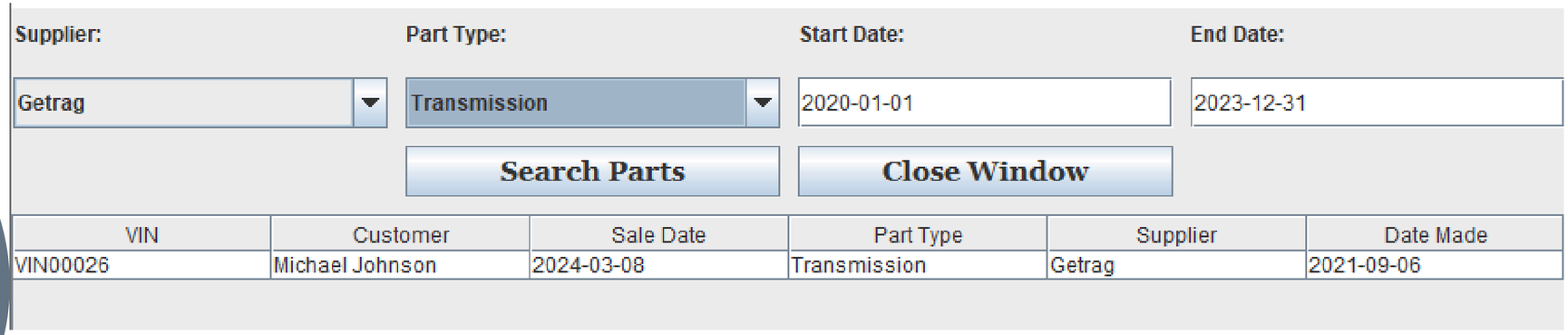


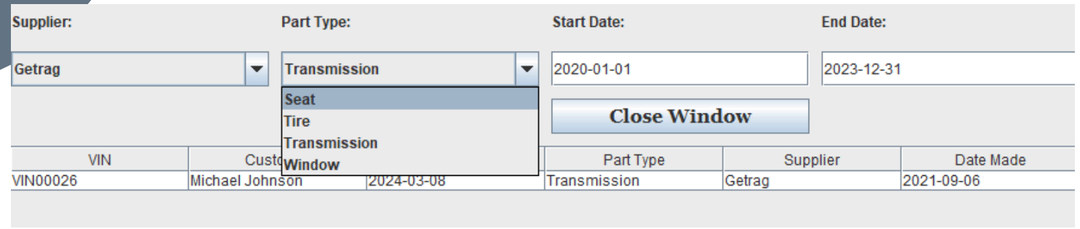
Brand	Units Sold
Honda	6
Nissan	5

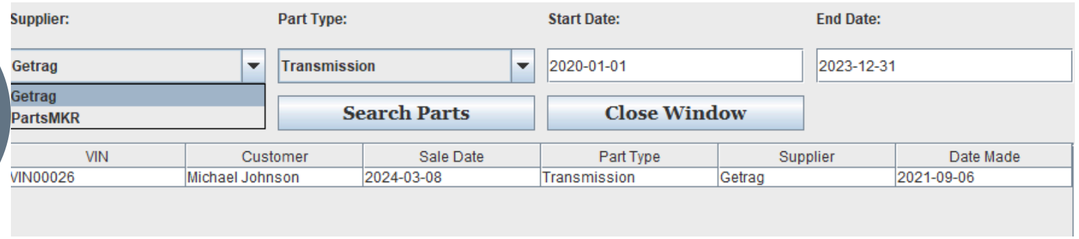


Month	Units Sold
4	3

INTERFACE – PARTS MENU



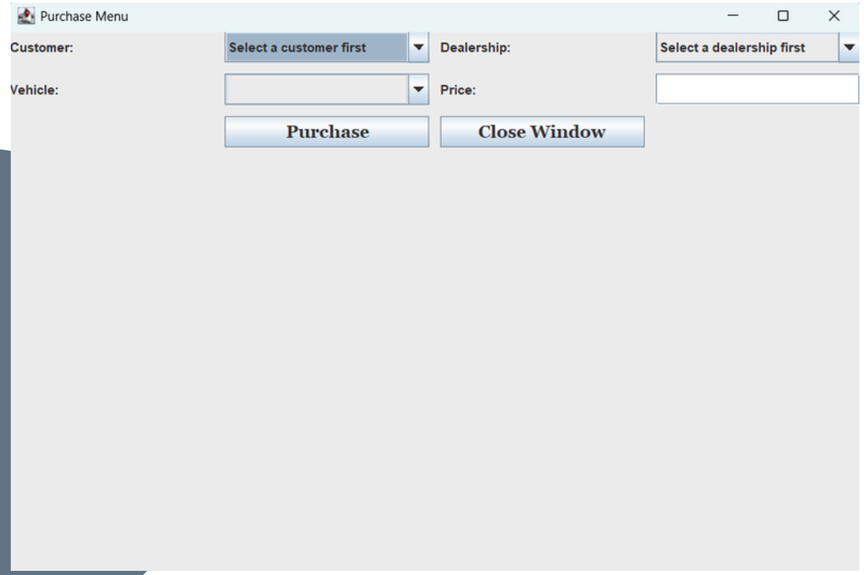


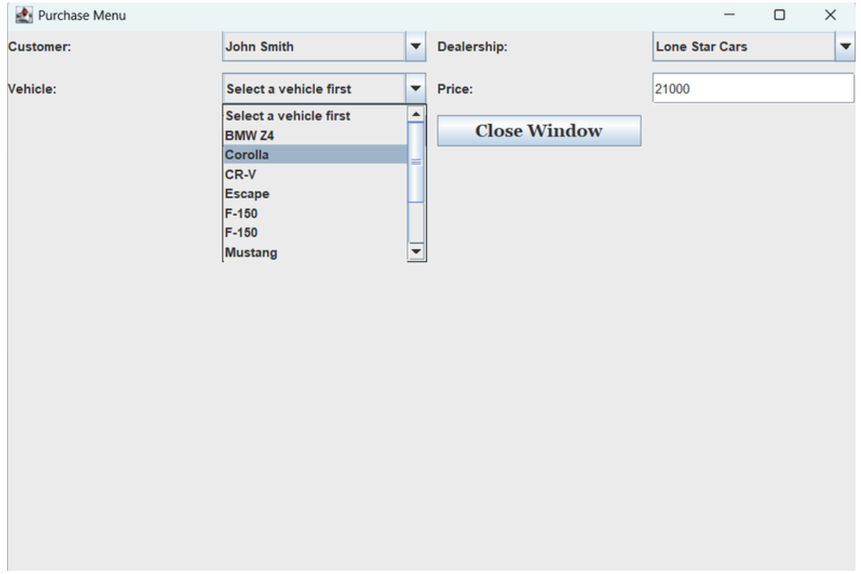


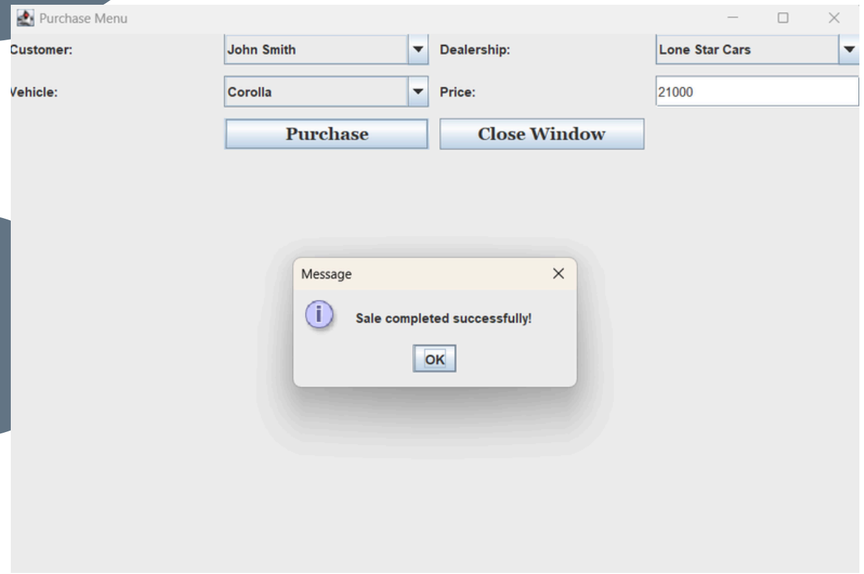
The **parts menu** allows the user to select a supplier and the part type that the supplier supplies. The user can then give a range for when that part was made, and any sale records with that date will be pulled. The report will list the following:

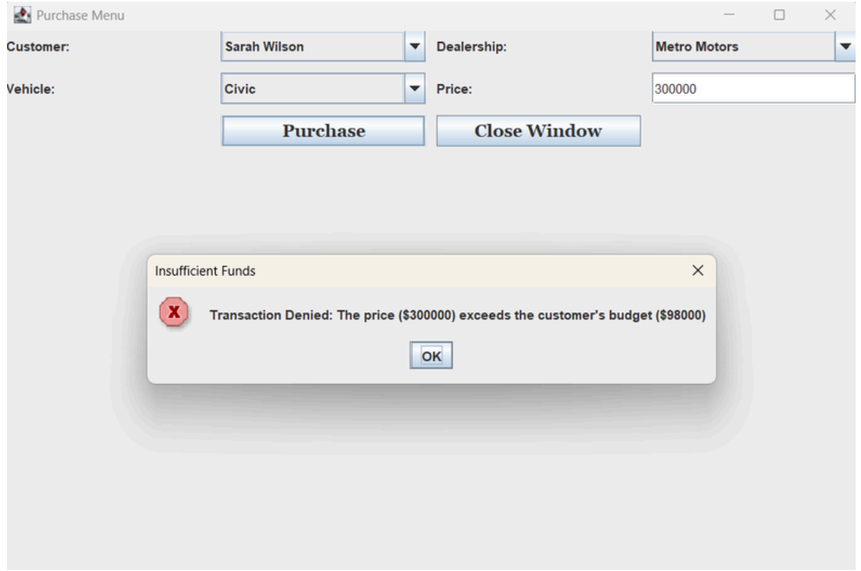
- VIN of the affected Vehicle
- Customer
- Sale Date
- Part Type
- Supplier
- Date made

INTERFACE – PURCHASE MENU









The **Purchase Menu** allows to select the customer who is buying a car, the dealership from which they want to buy, the cars available at the dealership, and the price the customer will pay.

If the price is lower than customer's income, the sale will completed successfully. Otherwise, transaction will be denied.

The End